

Han Zhou

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EMPLOYMENT

- **University of Pennsylvania** Philadelphia, PA, USA
AMCS/CMB Postdoctoral Fellow, Department of Mathematics Sep. 2024 -

EDUCATION

- **Shanghai Jiao Tong University** Shanghai, PRC
Ph.D., Mathematics, Advisor: Wenjun Ying. Sept. 2020 - Jun. 2024
- **Shanghai Jiao Tong University** Shanghai, PRC
B.S., Mathematics and Applied Mathematics Sept. 2016 - Jun. 2020

RESEARCH INTEREST

- **Scientific Computing & Numerical Analysis:** efficient numerical methods for interface problems and fluid-structure interaction
- **Mathematical Biology:** modeling and simulation of biological membranes, cells, and vesicles
- **PDE Analysis:** well-posedness for fluid-structure interaction and open-boundary systems

PAPERS AND PREPRINTS

1. Charles L. Epstein, Yoichiro Mori, **Han Zhou**. Bulk-surface coupled PDE with an open boundary. arXiv:2604.20798.
2. **Han Zhou**, Wenjun Ying. A correction function-based KFBI method for Brinkman interface problems. arXiv:2604.15509.
3. **Han Zhou**, Yuan-Nan Young, Yoichiro Mori. Modeling and simulation of open membranes in Stokes flow with mixed-dimensional coupling. *Multiscale Modeling & Simulation*, 24(2):474–500, 2026. DOI.
4. Pengsong Yin, Wenjun Ying, Yulin Zhang, **Han Zhou**. A kernel-free boundary integral method for elliptic interface problems on surfaces. *Journal of Computational Physics*, 562:115004, 2026. DOI.
5. **Han Zhou**, Shuwang Li, Wenjun Ying. An alternating direction implicit method for mean curvature flows. *Journal of Scientific Computing*, 101:65, 2024. DOI.
6. **Han Zhou**, Wenjun Ying. A correction function-based kernel-free boundary integral method for elliptic PDEs with implicitly defined interfaces. *Journal of Computational Physics*, 496:112545, 2024. DOI.
7. **Han Zhou**, Jiahe Yang, Wenjun Ying. A kernel-free boundary integral method for the nonlinear Poisson-Boltzmann equation. *Journal of Computational Physics*, 493:112423, 2023. DOI.
8. **Han Zhou**, Wenjun Ying. A Cartesian grid-based boundary integral method for moving interface problems. arXiv:2309.01068.
9. **Han Zhou**, Minsheng Huang, Wenjun Ying. ADI schemes for the heat equation on arbitrary 3D domains and their applications. arXiv:2309.00979.
10. **Han Zhou**, Wenjun Ying. A dimension splitting method for time dependent PDEs on irregular domains. *Journal of Scientific Computing*, 94(1):20, 2023. DOI.
11. Jiacheng Xu, Dan Hu, **Han Zhou**. A phase-field method for elastic mechanics with large deformation. *Journal of Computational Physics*, 471:111630, 2022. DOI.

TALKS

- 2026 SIAM Annual Meeting, Cleveland Jul. 6–10, 2026
- 2026 NCTS Workshop on Mathematics of Living Systems, Hsinchu May 20–22, 2026
- Kanazawa University - Penn Soft Matter / Applied Math Workshop, Philadelphia Mar. 23–25, 2026
- Mathematical Modeling, Computational Methods, and Biological Fluid Dynamics: Research and Training, Chicago Aug. 2025
- Numerical Analysis and PDE Seminar, Newark May 2025
- Fluid Mechanics and Waves Seminar, Newark Nov. 2024
- NCTS Seminar on PDE and Machine Learning, Online Oct. 2024
- The 10th International Congress on Industrial and Applied Mathematics, Tokyo Aug. 2023
- SICIAM Workshop on Recent Advances in Fast Algorithms, Shenzhen Aug. 2023
- Chinese Mathematical Society in Computational Mathematics, Nanjing Jul. 2023
- CSIAM Student Forum, Online Nov. 2022
- Shanghai Symposium on Scientific and Engineering Computing Methods, Shanghai Nov. 2021

HONORS AND SCHOLARSHIPS

- Good Teaching Awards for Spring 2025 (UPenn math) 2025
- National Scholarship 2023
- Academic Scholarship for Graduate Students 2020-2022
- Excellent Bachelor Thesis of School of Mathematical Sciences 2020

TEACHING

University of Pennsylvania

- **Instructor**, AMCS 6035, Numerical and Applied Analysis II, Spring 2026
- **Instructor**, AMCS 6025, Numerical and Applied Analysis I, Fall 2025
- **Instructor**, AMCS 6035, Numerical and Applied Analysis II, Spring 2025
- **Instructor**, AMCS 6025, Numerical and Applied Analysis I, Fall 2024

Shanghai Jiao Tong University

- **Teaching Assistant** Differential geometry (Fall 2023), Convex optimization (Spring 2022, Fall 2022), Scientific computing (Spring 2021, Fall 2021), Calculus (Fall 2020)

Graduate projects supervision(UPenn):

- Rui Xu (graduate student, AMCS) 2025
- JiYao Zhang (graduate student, AMCS) 2025

TECHNICAL SKILLS

Language: Mandarin, English

Computer programming C++, Python, MATLAB.

ACADEMIC SERVICE

- Organizer, weekly UPenn AMCS Seminar 2025-2026
- Organizer, weekly UPenn Mathematical Biology Seminar 2024-2025